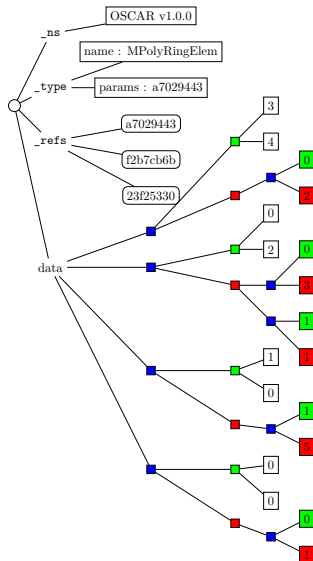


Serializing Algebraic Data

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Serializing Algebraic Data

Serialization: translate in-memory data to a form that can be stored or transmitted, then reconstructed on another process or machine.

Say we want to store:

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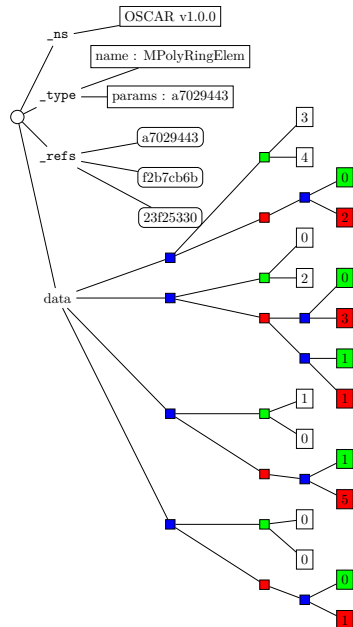
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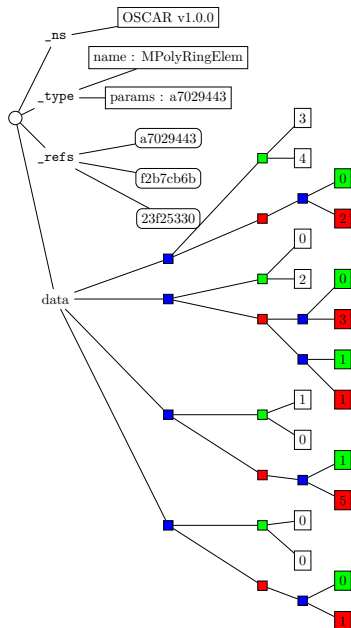
$$p = 2y^3z^4 + (a + 3)z^2 + 5ay + 1$$

- Is y considered a coefficient of z ?
- What is a ?
- How can we guarantee the objects behave as expected on load?
- Elements sharing a parent should have the same parent on load.

The mrdi File Format



The mrdi File Format



- ICMS 2024 [DV - Joswig - Lorenz].
- Initial focus on long term storage.
- JSON based.
- `_ns` pins semantics.
- `_type` subtree describes the data and it's algebraic context.
- Shared parents (e.g. the ring) stored once in `_refs`, addressed by UUID.
- Implementations for Macaulay2 (Doug Torrance), Magma (Jeroen Hanselman) and SageMath

Serialization Framework

Adding support for a new type means implementing three functions.

- `type_and_params` — returns the context needed to rebuild the object (e.g. its parent ring).
- `save_object` — writes the data subtree.
- `load_object` — reads the data subtree expects context to be provided.

Serialization Framework

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- `type_and_params` — returns the context needed to rebuild the object (e.g. its parent ring).
- `save_object` — writes the data subtree.
- `load_object` — reads the data subtree expects context to be provided.

- General framework handles container types.
- Entire sessions can be stored across multiple files.
- Coercion — loading files into a given algebraic context.

Serializers

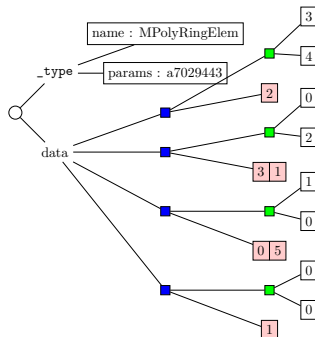
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function save_object(s::SerializerState, p::PolyRingElem)
    save_object(s, [
        (i - 1, c) for (i, c) in enumerate(coefficients(p))
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function save_object(s::SerializerState{IPCSerializer}, p::PolyRingElem)
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Serializers

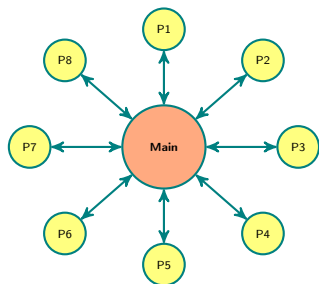
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- Selection of different serializers lets users alter subtrees.
- Parametrized SerializerState{S}
- Multiple dispatch on S selects the encoding per context
- Currently available: JSONSerializer, IPCSerializer, LPSerializer, MultiFileRef.

Multiprocessing



`OscarWorkerPool` — subtype of Julia's `AbstractWorkerPool`, Uses the `IPCSerializer`.

- Sends context to workers once, then reuses an `IPCSerializer` for every message.
- Julia's `remotecall` and `pmap` work directly with `OSCAR` types.
- Works with cluster managers (`SlurmClusterManager.jl`)
- **32 processes**, 150 GB: < 3 hours vs. > 24 hours (**1 process**) found degree-5 components of vanishing-ideal for $\sigma_4(\mathbb{P}^3 \times \mathbb{P}^3 \times \mathbb{P}^3)$.

Future Work

- Spin off the serialization code into a standalone Julia package.
- Add support for binary formats.

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Thank You!